Fission & Fusion

Exam code: 0625 & 0972

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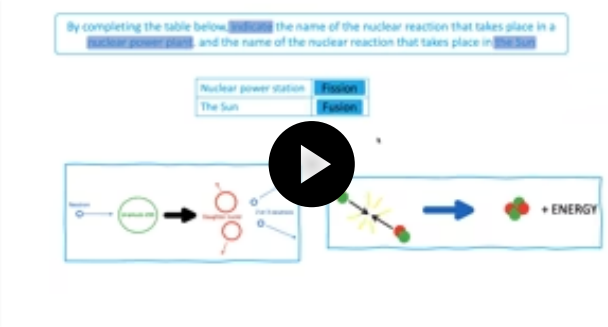

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Fission & fusion

- Nuclear fission & fusion are **nuclear reactions** that change the nucleus of an atom to produce high amounts of energy from the energy **stored** in the nucleus of an atom

Nuclear fission

- Nuclear fission is defined as:

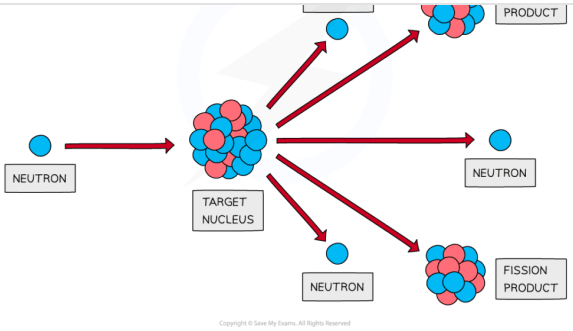
The splitting of a large, unstable nucleus into two smaller nuclei

- During fission:
  - A neutron collides with an unstable nucleus
    - The neutron and the nucleus are the **reactants**
  - The nucleus splits into **two smaller nuclei** (called daughter nuclei) and **two or three neutrons**
    - The daughter nuclei and the neutrons are the **products** of the reaction
  - Gamma rays are also emitted

Nuclear fission process

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



A neutron is fired into the target nucleus, causing it to split

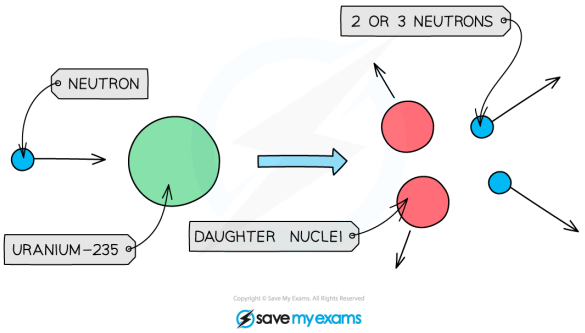
Nuclear fission nuclide equations

- An example of a nuclide equation for the fission of uranium-235 is:

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{92}_{36}\text{Kr} + ^{141}_{56}\text{Ba} + 3\,^{1}_{0}\text{n} + \text{energy}$$

- Where:
  - $^{235}_{92}\text{U}$ is an unstable isotope of uranium
  - $^{1}_{0}\text{n}$ is a neutron
  - $^{92}_{36}\text{Kr}$ is an unstable isotope of krypton
  - $^{141}_{56}\text{Ba}$ is an unstable isotope of barium

Nuclear fission of uranium-235



Large nuclei can decay by fission to produce smaller nuclei and neutrons with a lot of kinetic energy

Nuclear fission mass and energy values

- **Energy is conserved** in a nuclear fission reaction
- In the example:

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{92}_{36}\text{Kr} + ^{141}_{56}\text{Ba} + 3\,^{1}_{0}\text{n} + \text{energy}$$

- The sum of the nucleon (top) numbers of the reactants (left-hand side) is equal to the sum of the nucleon numbers of the products (right-hand side):

$$235 + 1 = 92 + 141 + (3 \times 1)$$

- The products of fission move away very **quickly**
  - During a fission reaction, energy is transferred from **nuclear energy** store of the parent nucleus to the kinetic energy store of the products
- The mass of the products is fractionally **less** than the mass of the original nucleus
  - This is because the remaining mass has been converted into **energy,** which is released during the fission process
- Large isotopes with a large nucleon number, such as **uranium** and **plutonium,** both undergo fission and are used as fuels in nuclear power stations

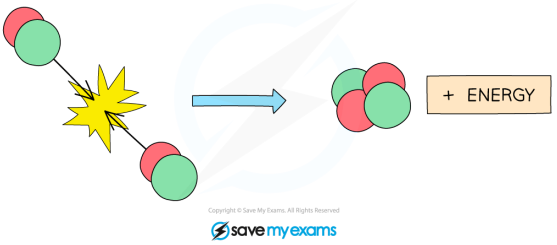
Nuclear fusion

- Nuclear fusion is defined as:

When two light nuclei join to form a heavier nucleus

- Stars use nuclear fusion to produce **energy**
  - In most stars, hydrogen nuclei (**light nuclei**) are fused together to form a helium nucleus (**heavier nucleus**) and massive amounts of **energy** is produced

Nuclear fusion of hydrogen



Two hydrogen nuclei fuse to form a helium nucleus

- Nuclear fusion requires extremely high **temperature** and **pressure**
  - So fusion is very hard to reproduce on Earth

Nuclear fusion nuclide equations

- An example of a nuclide equation for fusion is:

$$^{2}_{1}\text{H} + ^{1}_{1}\text{H} \rightarrow ^{3}_{2}\text{He} + \text{energy}$$

- Where:
  - $^{2}_{1}\text{H}$ is deuterium (isotope of hydrogen with 1 proton and 1 neutron)
  - $^{1}_{1}\text{H}$ is hydrogen (with one proton)
  - $^{3}_{2}\text{He}$ is an isotope of helium (with two protons and one neutron)

Nuclear fusion mass and energy values

- Therefore, the mass of the product (fused nucleus) is fractionally **less** than the mass of the two original nuclei (reactants)
  - The remaining mass has been converted into the **energy** released when the nuclei fuse
- The amount of **energy released** during nuclear fusion is **huge**:
  - The energy from 1 kg of hydrogen that undergoes fusion is equivalent to the energy from burning about 10 million kilograms of coal

Worked Example

A nuclide equation for nuclear fission is stated as:

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{96}_{37}\text{Rb} + ^{137}_{55}\text{Cs} + N^{1}_{0}\text{n}$$

Calculate the number of neutrons, *N* emitted in this reaction.

Answer:

Step 1: Calculate the sum of the nucleon numbers of the reactants

- The reactants are on the left-hand side of the equation
- The nucleon numbers are the top numbers in the nuclide notation

$$235 + 1 = 236$$

Step 2: Calculate the sum of the nucleon numbers of the products

- The products are on the right-hand side of the equation

$$96 + 137 + (N \times 1) = 233 + N$$

Step 3: Equate the total nucleons of the reactants and products

$$236 = 233 + N$$

Step 4: Rearrange for the number of neutrons, N

$$N = 236 - 233 = 3$$

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